PCA theorem Slide 6

1) Consider $z_1 = \varphi_1^T X$ $\quad \varphi_1 : 1\times p,\ X : p\times 1$

find φ_1 such that $\operatorname{var}(\varphi_1^T X) = \varphi_1^T \Sigma \varphi_1$ is maximized

apply Fact 2(a) with $B = \Sigma$, then

$\varphi_1 = e_1$ and $\max \varphi_1^T \Sigma \varphi_1 = \lambda_1$

(Note: we can always ensure $\varphi_1^T \varphi_1 = 1 = \sum_{j=1}^{p} \varphi_{1j}^2$)

What abut z_r, $r > 1$? prove by induction as below.

Assume we have $z_1, \dots, z_k$ with $\varphi_1 = e_1, \dots, \varphi_k = e_k$

we want to find z_{k+1} such that

$\operatorname{Var}(z_{k+1}) = \operatorname{Var}(\varphi_{k+1}^T X)$ is maximized and $\operatorname{Cov}(z_{k+1}, z_r) = 0$ for $r < k+1$

where

$\operatorname{Var}(z_{k+1}) = \varphi_{k+1}^T \Sigma \varphi_{k+1}$

$\operatorname{Cov}(z_{k+1}, z_r) = \varphi_{k+1}^T \Sigma \varphi_r$

Since $\varphi_r = e_r$, we have $\Sigma e_r = \lambda_r e_r$

therefore $\operatorname{Cov}(z_{k+1}, z_r) = \varphi_{k+1}^T \cdot \lambda_r e_r$

$= \lambda_r \varphi_{k+1}^T e_r = 0$

i.e $\varphi_{k+1}^T e_r = 0$

so we want $\varphi_{k+1}^T e_r = 0$ and $\varphi_{k+1}^T \Sigma \varphi_r$ is maximized.

By Fact 2(b) with $\Sigma = B$

we have $\varphi_{k+1} = e_{k+1}$ (and $\varphi_{k+1}^T \varphi_r = 1$ by definition)

2) $\operatorname{Var}(z_r) = \varphi_r^T \Sigma \varphi_r$

$= e_r^T \Sigma e_r = e_r^T \lambda_r e_r = \lambda_r e_r^T e_r = \lambda_r$

3) $\operatorname{Cov}(z_j, z_r) = \operatorname{Cov}(e_j^T X, e_r^T X) = e_j^T \Sigma e_r$

$= e_j^T \lambda_r e_r = \lambda_r e_j^T e_r = 0$

$e_j^T e_r = 0$ (orthogonal)

4) $\sum_{j=1}^{p} Var(X_j)$ = sum of diagonal elements of Σ

$= trace(\Sigma) = trace(P \Lambda P^T)$

$\downarrow$

By Fact 1

$= trace(\Lambda P P^T) = trace(\Lambda) = \sum_{j=1}^{p} \lambda_j$

Note $trace(AB) = trace(BA)$

5) $\frac{Var(Z_j)}{\sum Var(Z_j)} = \frac{\lambda_j}{\sum_{j=1}^{p} \lambda_j}$

6) $\rho_{Z_r, X_k} = \frac{Cov(Z_r, X_k)}{\sqrt{Var(Z_r)} \cdot \sqrt{Var(X_k)}}$

$= \frac{Cov(e_{rk} X_k, X_k)}{\sqrt{\lambda_r} \cdot \sqrt{Var(X_k)}} = \frac{\lambda_r \cdot e_{rk}}{\sqrt{\lambda_r} \cdot \sqrt{Var(X_k)}} = \frac{e_{rk} \sqrt{\lambda_r}}{\sqrt{Var(X_k)}}$

where $Cov(X_k, Z_r) = Cov(a_k^T X, e_r^T X)$

$= a_k^T \Sigma e_r$

$= a_k^T \lambda e_r$

$= \lambda a_k^T e_r$

$= \lambda e_{rk}$

$a_k^T = (0, \cdots 0, \underset{\downarrow k}{1}, 0, \cdots, 0)$

$a_k^T X = X_k$